



**FACULTY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND COMPUTER
ENGINEERING**

**ENEE5306
PROTECTION IN POWER SYSTEM**

Assignment 1

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Abstract

This assignment aims to provide knowledge about the EATP simulation app and understand the tools to do a simulation for a power system circuit, perform the normal operation, and short operation at each bus, and add the current transformer, and see the voltage profile for the buses in each operation, also the function of Fuse

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Phase 1

Settings

Transformers

15 MVA IEC Liquid-Fill Other 65 C

161 33 kV

Voltage Rating

	kV	FLA	Nominal Bus kV
Prim.	161	53.79	161
Sec.	33	262.4	33

Other 65

Power Rating

MVA

Rated 15

Other 65

Derated 15

% Derating 0

MFR

Alert - Max

MVA

15

☒ Derated MVA

☐ User-Defined

Installation

Altitude 1000 m

Ambient Temp. 30 °C

Type / Class

Type	Sub Type	Class	Temp. Rise
Liquid-Fill	Other	Other	65

Figure 2: main transformer

2.5 MVA IEC Liquid-Fill Other 65 C

33 0.4 kV

Voltage Rating

	kV	FLA	Nominal Bus kV
Prim.	33	43.74	33
Sec.	0.4	3608	0.4

Other 65

Power Rating

MVA

Rated 2.5

Other 65

Derated 2.5

% Derating 0

MFR

Alert - Max

MVA

2.5

☒ Derated MVA

☐ User-Defined

Installation

Altitude 1000 m

Ambient Temp. 30 °C

Type / Class

Type	Sub Type	Class	Temp. Rise
Liquid-Fill	Other	Other	65

Figure 1: distribution transformer

The transformer data depends on the load, where the mean bus will be loaded to a maximum 50%, and the distribution will be 80% of the rating, which was a little higher than what is needed. Reduce the voltage drop, and the X/R ratio was set at a typical value

Transmission line

Transmission Line Editor - Line9

Sag & Tension Ampacity Compensation Rel

Info Parameter Configuration Grouping

Impedance (per phase)

	R - T1	R - T2	X	Y
Pos.	0.2	0.2	0.646	0
Neg.	0.2	0.2	0.646	0
Zero	0.2	0.2	0.646	0

Figure 3: Transmission line data

Load

Lumped Load Editor - Lump2

Info Nameplate Short-Circuit Dyn Model Time Domain Reliability Remarks Comment

1.11 MVA 0.4 kV (80% Motor 20% Static)

Model Type
Conventional

Rated kV
0.4

Calculator...

Ratings

MVA	MW	Mvar	% PF	Amp
1.11	1.021	0.435	92	1602

Load Type

Constant kVA
0 80 % 100

Constant Z
100 20 % 0

Figure 4:load data

Power Grid Editor - U1

Info Rating Short Circuit Time Domain Harmonic Reliability Energy Price Remarks Comment

161 kV Swing

Grounding

SC Rating

MVAAsc	X/R	kAsc
4019	15	14.412

SC Impedance (100 MVA)

	% R	% X
Pos.	0.16551	2.48267
Neg.	0.16551	2.48267

Figure 5: power grid data

The X/R ratio of the grid can be between 15-20, and the impedance will be automatically calculated

Part A normal operation

Without Transformer

Normal

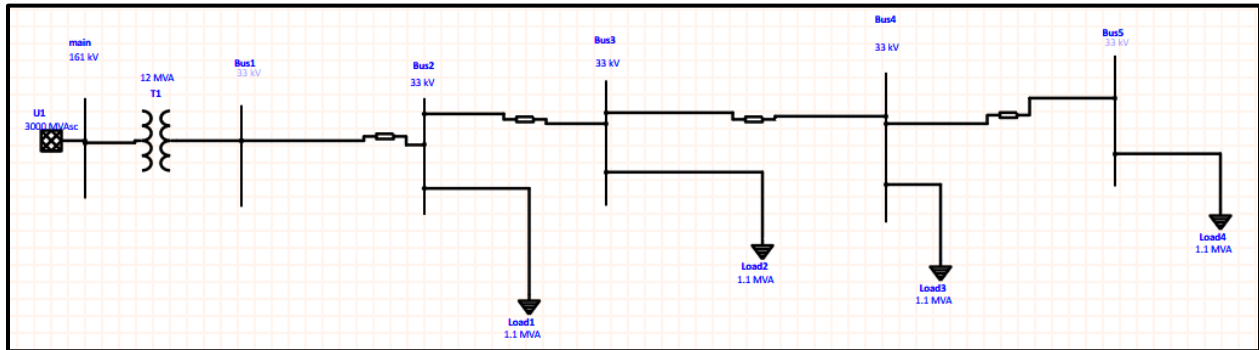


Figure 6 Normal operation without a transformer

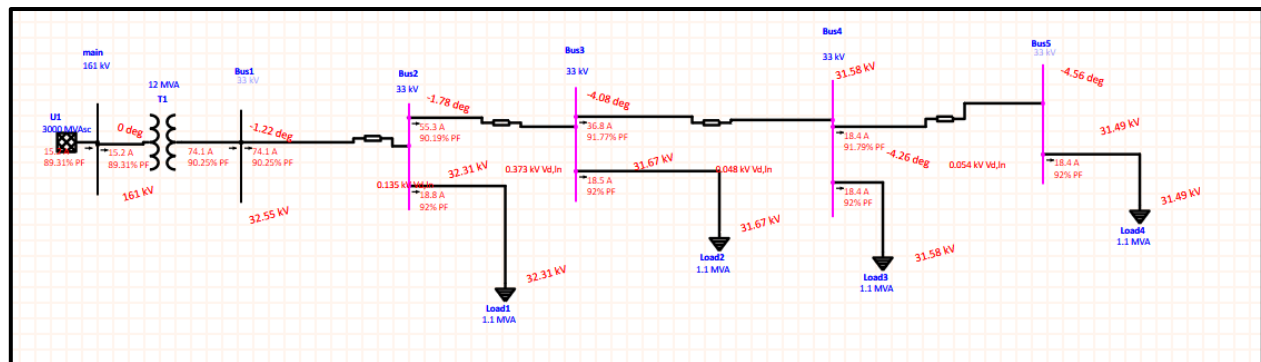


Figure 7 power flow (without transformer)

The system was implemented without the transformer to see the effect of the transformer on the network and compare the results, A power system's voltage decreases as the number of buses increase because electricity travels farther through more transformers and transmission lines, each of which contributes to a minor voltage loss that lowers the system voltage

Bus Loading Summary Report														
Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic		MVA	% PF	Amp	Percent Loading
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar				
Bus1	33.000										4.265	91.5	75.4	
Bus2	33.000				0.980	0.418					4.242	91.8	75.4	
Bus3	33.000				0.972	0.414					3.164	91.9	56.5	
Bus4	33.000				0.967	0.412					2.102	92.0	37.6	
Bus5	33.000				0.965	0.411					1.049	92.0	18.8	
main	161.000										4.310	90.8	15.5	
* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).														
# Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).														

Figure 8:buses data (without transformer)

Branch Losses Summary Report									
Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd
	MW	Mvar	MW	Mvar	kW	kvar	From	To	% Drop in Vmag
Line10	1.935	0.833	-1.933	-0.825	2.5	8.2	98.0	97.8	0.26
Line11	0.965	0.413	-0.965	-0.411	0.6	2.1	97.8	97.6	0.13
Line8	3.904	1.717	-3.894	-1.684	10.2	33.1	98.9	98.4	0.52
Line9	2.914	1.266	-2.908	-1.248	5.7	18.5	98.4	98.0	0.39
T1	-3.904	-1.717	3.914	1.805	10.3	88.0	98.9	100.0	1.06
					29.5	149.8			

* This Transmission Line includes Series Capacitor.

Figure 9: branch losses without transformer

The first table gives the voltage and power consumed by each of the electricity connection buses. The second table shows the amount of voltage drop and power lost as electricity travels through various lines. Together, they show where the energy is lost and how effectively the power system is carrying its load.

	Bus ID	Nominal kV	Amp Rating	Voltage	MW Loading	Amp Loading	
1	Bus1	33	0	98.94	3.904	75.41	
2	Bus2	33	0	98.42	3.894	75.41	
3	Bus3	33	0	98.03	2.908	56.47	
4	Bus4	33	0	97.77	1.933	37.61	
5	Bus5	33	0	97.64	0.965	18.79	
6	main	161	0	100	3.914	15.46	

Figure 10: busses voltages (without transformer)

	ID	Type	kW Flow	kvar Flow	Amp Flow	% Loading
1	Line8	Line	3904	1716.8	75.41	
2	Line9	Line	2913.5	1266.2	56.47	
3	Line10	Line	1935.3	833.3	37.61	
4	Line11	Line	965.4	413	18.79	
5	T1	Transf. 2W	3914.3	1804.8	15.46	28.7

Figure 11: lines data (without transformer)

	ID	Unit	Rated kV	kW	kvar	Amp	% PF
1	Load1	VA	33	980.3	417.6	18.94	92
2	Load2	VA	33	972.5	414.3	18.87	92
3	Load3	VA	33	967.3	412.1	18.82	92
4	Load4	VA	33	964.8	411	18.79	92

Figure 12:load data

This output provides three tables of a power system. The first table shows voltages and power consumption at different connection points called buses. The second table shows power flow and current in transmission lines, representing how much each line is being used. The third table specifies individual loads with power consumption and power factor.

fault

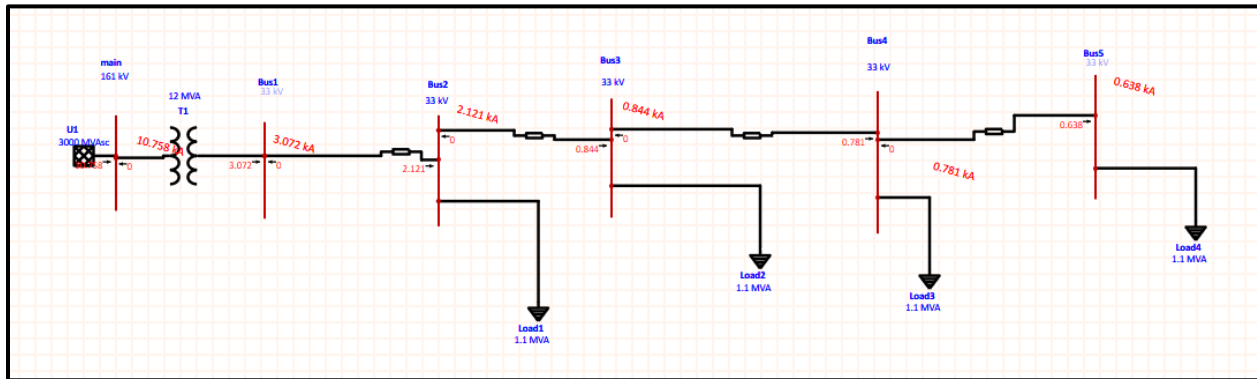


Figure 13: Fault operation without Transformer

The fault current on the main bus will be the highest since the fault current depends on the load that is connected to the buses. Current is high at the source and starts to decrease through the radial system. It dips to 3.81 kA at Bus1, 2.793 kA at Bus2, and continues to decrease until it is 1.546 kA at Bus5. This is due to the impedance of cables and transformers

With Transformer (modified)

Normal

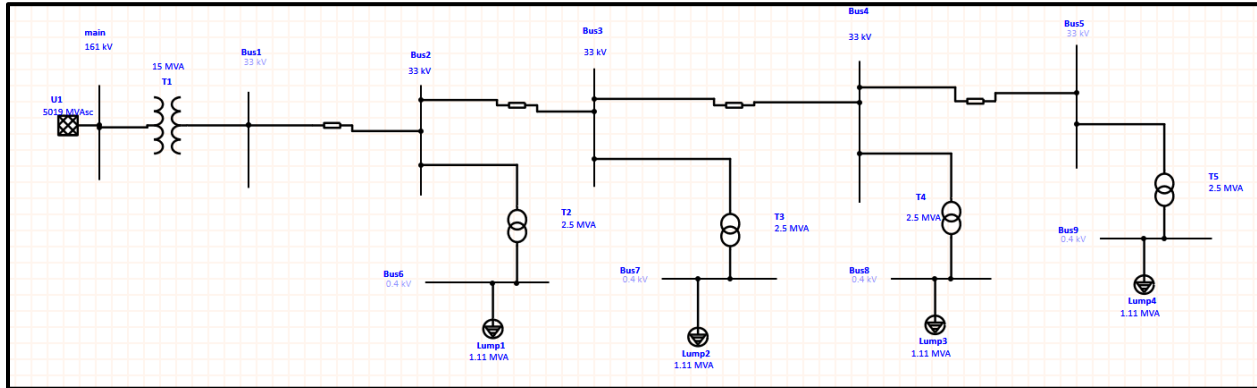


Figure 14: network system modified

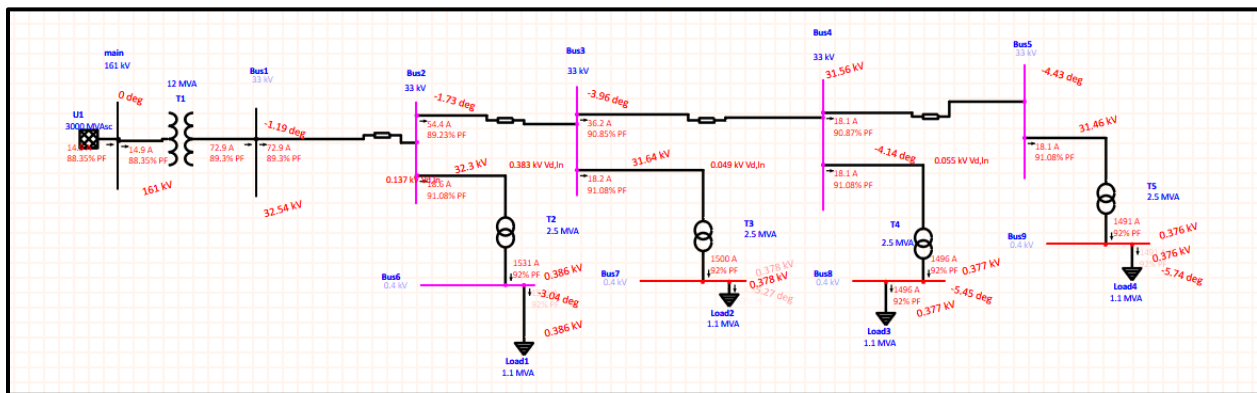


Figure 15: power flow in the modified

The network was designed and implemented using ETAP, and it has been simulated in normal mode to see the efficiency of the system and to detect the effect of the fault when it happens, also to determine the proper protection device and see the impact of it on the system. This network consists of five sockets, aside from a 1 MW load, aside from the transmission line, where is the impedance of the transmission line equal to $R = 0.2066$ and $X = 0.64876$.

	Bus ID	Nominal kV	Amp Rating	Voltage	MW Loading	Amp Loading
1	Bus1	33	0	98.85	4.066	79.41
2	Bus2	33	0	98.75	4.064	79.41
3	Bus3	33	0	98.03	3.038	59.65
4	Bus4	33	0	97.74	2.023	39.8
5	Bus5	33	0	97.55	1.01	19.91
6	Bus6	0.4	0	97.2	1.01	1630
7	Bus7	0.4	0	96.48	1.007	1638
8	Bus8	0.4	0	96.19	1.006	1641
9	Bus9	0.4	0	95.99	1.005	1643
10	main	161	0	100	4.078	16.28

Figure 16:buses voltage (modified network)

	ID	Type	kW Flow	kvar Flow	Amp Flow	% PF
1	Line8	Line	4066.1	1895.9	79.41	90.63
2	Line9	Line	3049	1428.1	59.65	90.56
3	Line10	Line	2025.6	932.8	39.8	90.83
4	Line11	Line	1011.5	463.2	19.91	90.92
5	T1	Transf. 2W	4077.6	1993.4	16.28	89.84
6	T2	Transf. 2W	1015.2	461.7	19.76	91.03
7	T3	Transf. 2W	1012.3	460.8	19.85	91.02
8	T4	Transf. 2W	1011.2	460.4	19.89	91.01
9	T5	Transf. 2W	1010.5	460.1	19.91	91.01

Figure 17: branches voltages (modified network)

Voltage profile

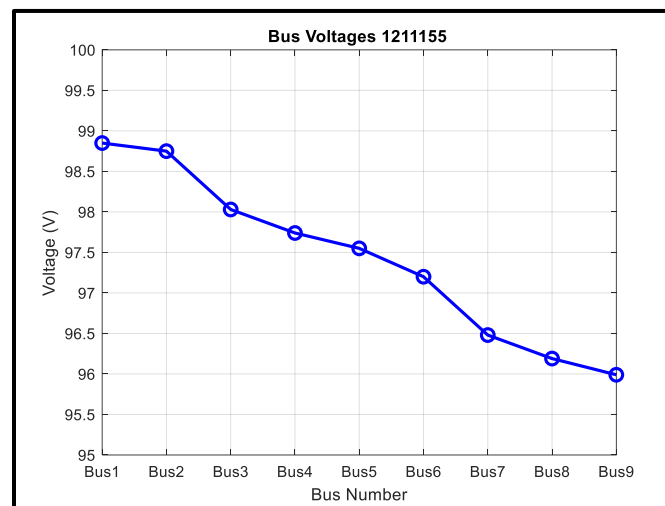


Figure 18: voltage profile

The voltage will decrease as the number of buses increases. The voltage profile shows a Smooth voltage drop from Bus 1 to Bus 9. Starting at 99 % of the voltage, it starts to decrease as power flows downstream through the network. This decline is expected due to line impedance and load consumption with buses. Bus 9, the voltage reaches around 95.8 %, indicating a typical voltage drop in radial distribution.

Branch Losses Summary Report									
Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
Line10	2.024	0.932	-2.021	-0.922	2.9	9.2	97.8	97.6	0.28
Line11	1.011	0.462	-1.010	-0.460	0.7	2.3	97.6	97.4	0.14
Line8	4.067	1.911	-4.055	-1.874	11.4	36.8	98.8	98.3	0.57
Line9	3.042	1.413	-3.036	-1.392	6.4	20.7	98.3	97.8	0.43
T1	-4.067	-1.911	4.078	2.009	11.5	97.8	98.8	100.0	1.16
T2	1.013	0.461	-1.008	-0.429	5.3	31.6	98.3	96.7	1.55
T3	1.012	0.461	-1.006	-0.429	5.3	31.8	97.8	96.3	1.56
T4	1.011	0.460	-1.005	-0.428	5.3	31.9	97.6	96.0	1.56
T5	1.010	0.460	-1.005	-0.428	5.3	32.0	97.4	95.9	1.56
					54.1	294.3			

Figure 19: branch losses modified network

LOAD FLOW REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR	
ID	kV	% Mag	Ang	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap	
Bus1	33.000	98.839	-1.1	0.000	0.000	0.000	0.000	Bus2	4.067	1.911	79.5	90.5		
								main	-4.067	-1.911	79.5	90.5		
Bus2	33.000	98.270	-1.4	0.000	0.000	0.000	0.000	Bus1	-4.055	-1.874	79.5	90.8		
								Bus3	3.042	1.413	59.7	90.7		
								Bus6	1.013	0.461	19.8	91.0		
Bus3	33.000	97.845	-1.7	0.000	0.000	0.000	0.000	Bus2	-3.036	-1.392	59.7	90.9		
								Bus4	2.024	0.932	39.8	90.8		
								Bus7	1.012	0.461	19.9	91.0		
Bus4	33.000	97.562	-1.9	0.000	0.000	0.000	0.000	Bus3	-2.021	-0.922	39.8	91.0		
								Bus5	1.011	0.462	19.9	90.9		
								Bus8	1.011	0.460	19.9	91.0		
Bus5	33.000	97.421	-2.0	0.000	0.000	0.000	0.000	Bus4	-1.010	-0.460	19.9	91.0		
								Bus9	1.010	0.460	19.9	91.0		
Bus6	0.400	96.718	-2.8	0.000	0.000	1.008	0.429	Bus2	-1.008	-0.429	1635.1	92.0		
Bus7	0.400	96.288	-3.1	0.000	0.000	1.006	0.429	Bus3	-1.006	-0.429	1639.7	92.0		
Bus8	0.400	96.002	-3.3	0.000	0.000	1.005	0.428	Bus4	-1.005	-0.428	1642.7	92.0		
Bus9	0.400	95.859	-3.4	0.000	0.000	1.005	0.428	Bus5	-1.005	-0.428	1644.2	92.0		
* main	161.000	100.000	0.0	4.078	2.009	0.000	0.000	Bus1	4.078	2.009	16.3	89.7		

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)
Indicates a bus with a load mismatch of more than 0.1 MVA

Figure 20: load flow in modified circuit

As the number of buses increases, the voltage drop decreases since the power will travel longer distances in wires. These wires add resistance, which causes a voltage drop higher on the buses that are too far from the source. The Branch Losses provides power losses (kW/kvar) and voltage drops across lines and transformers. The Load Flow Report presents voltage levels, generation, and load for all buses, along with power flow between buses. Voltage levels are within acceptable range. Overall system losses are 54.1 kW and 294.3 kvar, which indicates modest efficiency.

Part B: Fault analyses

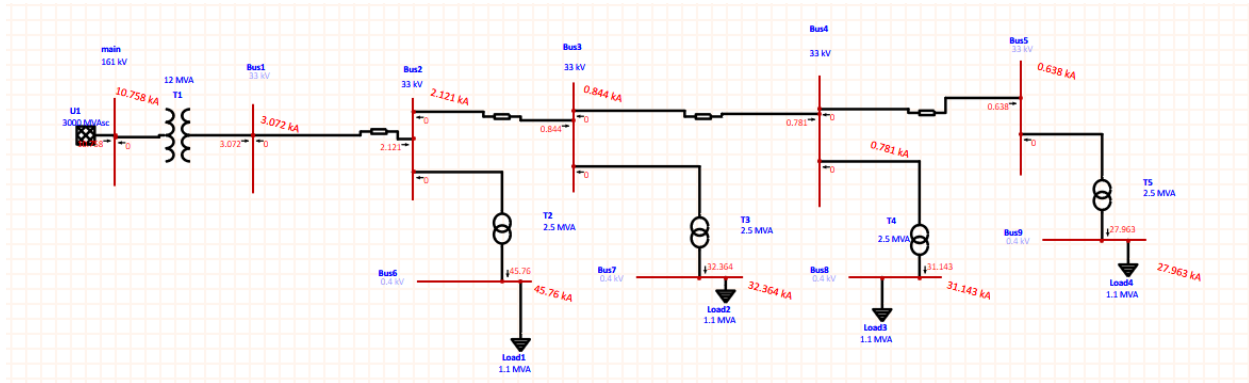


Figure 21: fault in the modified circuit

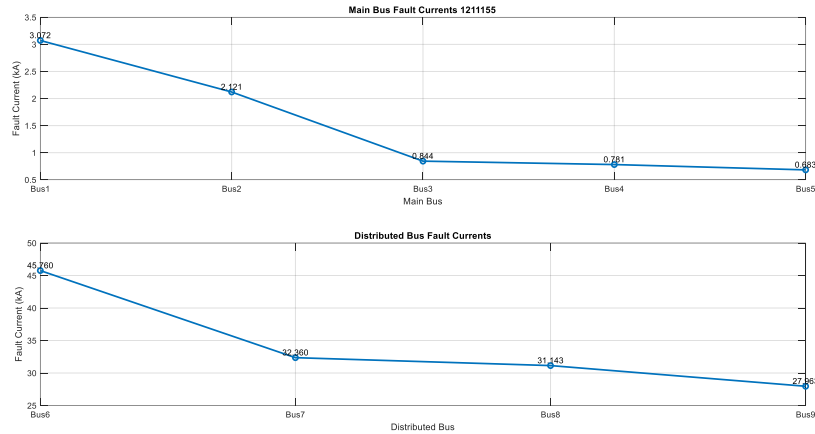


Figure 22: fault current at the buses

The fault was implemented at each bus to see the fault current that goes through each bus to determine the appropriate protection device, and to coordinate the protection devices with each other. The above two curves show the fault current at each bus, the Transformers step voltage from 33 kV down to 0.4 kV, enabling efficient, safe delivery to low-voltage loads. By confining the voltage drop and stepping up current only near the load, they minimize I^2R losses in upstream lines. This reduces transmission losses, improves voltage regulation, and prevents fault back-feed to the system. Strategic placement of these transformers improves overall system efficiency and reliability.

Table 1: Currents in normal and fault operation

Bus name	Normal current	Fault current
BUS1	72.4	3.072k
BUS2	54.7	2.121k
BUS3	36	0.844k
BUS4	18.1	0.781k
BUS5	18.1	0.683k
BUS6	1530	45.76k
BUS7	1500	32.36k
BUS8	1496	31.143k
BUS9	1491	27.963k

Branch Losses Summary Report									
Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
Line10	2.026	0.933	-2.023	-0.924	2.9	9.2	98.0	97.7	0.29
Line11	1.011	0.463	-1.010	-0.460	1.0	3.1	97.7	97.6	0.19
Line8	4.066	1.896	-4.064	-1.890	2.0	6.1	98.8	98.8	0.10
Line9	3.049	1.428	-3.038	-1.394	11.0	34.5	98.8	98.0	0.72
T1	-4.066	-1.896	4.078	1.993	11.5	97.5	98.8	100.0	1.15
T2	1.015	0.462	-1.010	-0.430	5.2	31.5	98.8	97.2	1.55
T3	1.012	0.461	-1.007	-0.429	5.3	31.7	98.0	96.5	1.55
T4	1.011	0.460	-1.006	-0.429	5.3	31.9	97.7	96.2	1.56
T5	1.010	0.460	-1.005	-0.428	5.3	31.9	97.6	96.0	1.56
					49.6	277.5			
* This Transmission Line includes Series Capacitor.									

Figure 23:branches losses

The branch losses and the fault current in each bus are shown above, The fault current will decrease depending on the load that be connected to each bus. For example, the fault at the main bus will be the highest since all the loads and the transformer are connected to it

Short-Circuit Summary Report

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I''k	ip	Ib sym	Ib asym	Idc	Ik
Bus1	33.000	Bus1	Bus					3.072	7.110				3.072
Bus2	33.000	Bus2	Bus					2.121	4.833				2.121
Bus3	33.000	Bus3	Bus					0.844	2.108				0.844
Bus4	33.000	Bus4	Bus					0.781	1.913				0.781
Bus5	33.000	Bus5	Bus					0.638	1.564				0.638
Bus6	0.400	Bus6	Bus					45.760	104.415				45.760
Bus7	0.400	Bus7	Bus					32.364	77.103				32.364
Bus8	0.400	Bus8	Bus					31.143	73.615				31.143
Bus9	0.400	Bus9	Bus					27.963	66.386				27.963
main	161.000	main	Bus					10.758	16.261				10.758

Figure 24: short circuit data

This report shows the 3-phase fault currents for all buses in the system. Fault current starts high at Bus1 (3.81 kA) and decreases to Bus5 (1.546 kA) according to system impedance. Breaker ratings are designed to withstand the fault levels that are expected. The high losses and voltage drops are in low-voltage branches, particularly T2-T5. Short-circuit currents are significantly higher at 0.4 kV buses because of low impedance and closeness to loads. Transformers restrict fault propagation upstream and enhance protection and system security.

Phase 2

Part C: Current transformer

By hand calculation

In this part, the hand calculation is needed to find the value for the current tap setting and the time delay setting for each circuit breaker to coordinate between breakers and to be used in the

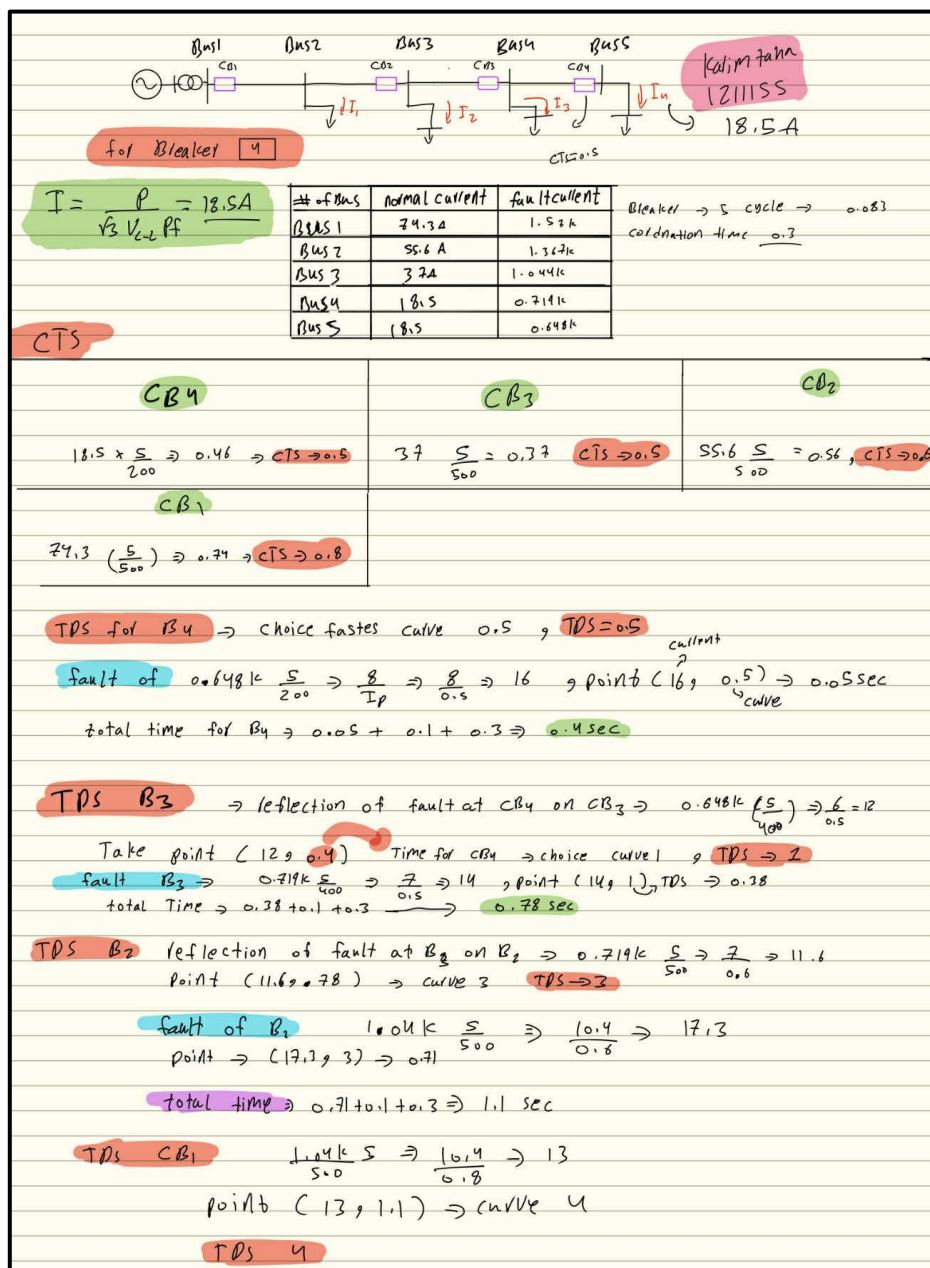


Figure 26: by hand calculation for really

Table 2 settings for relays

Circuit breaker	CTS	TDS
CB1	0.8	4
CB2	0.6	3
CB3	0.5	2
CB4	0.5	0.5

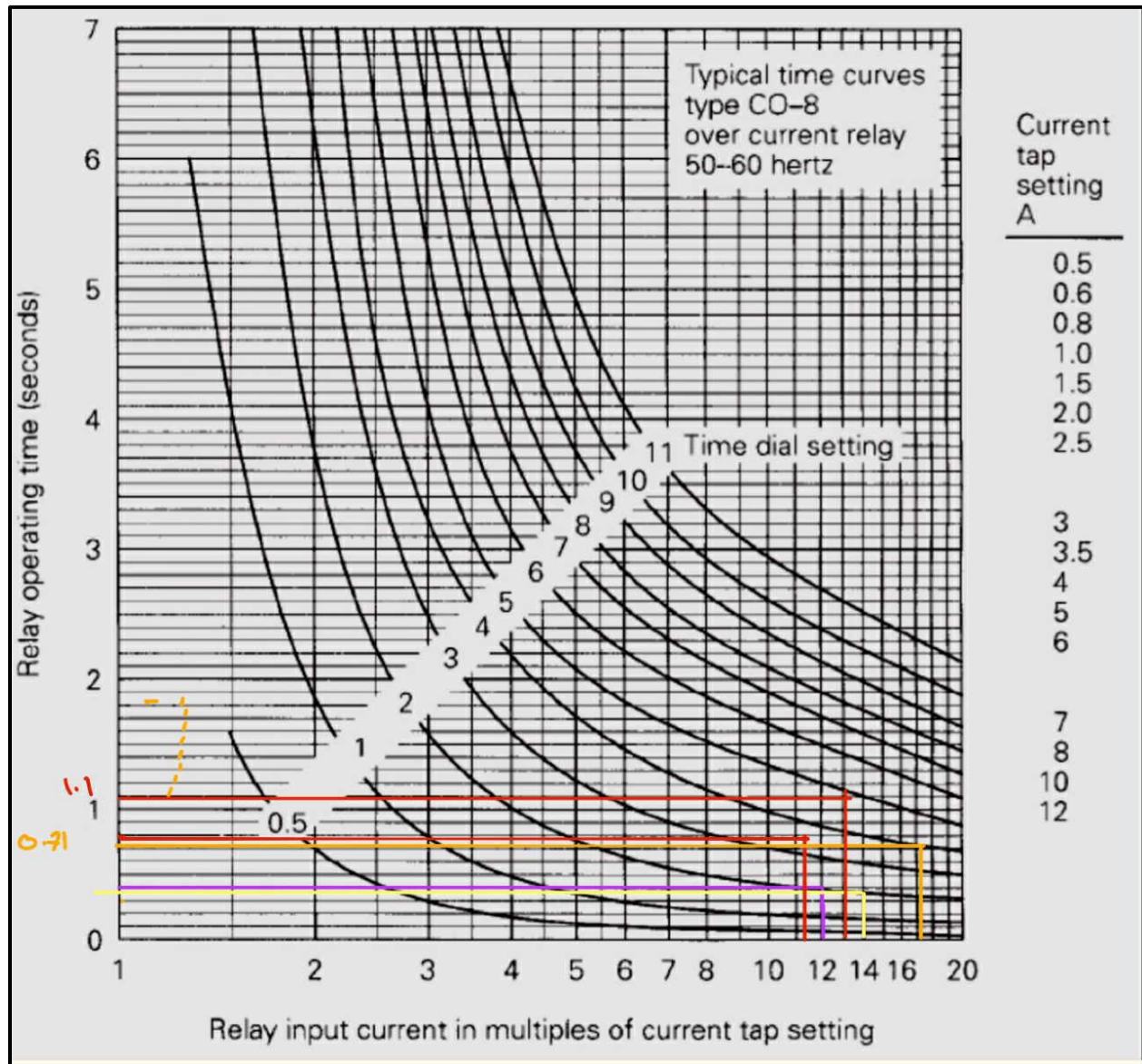
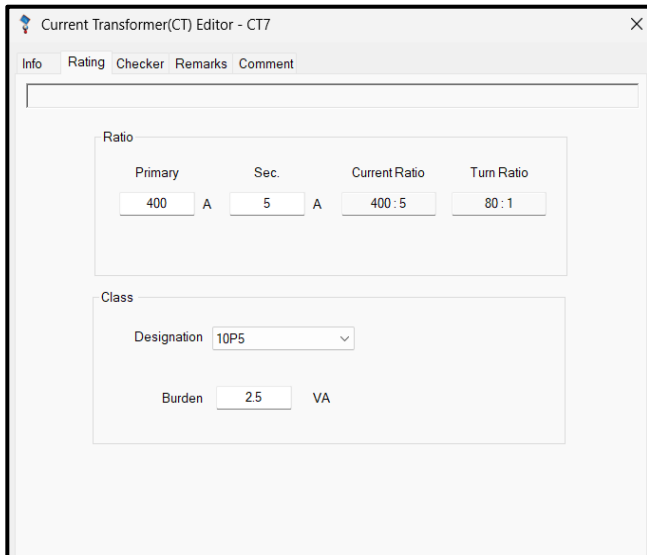


Figure 27: TDS points for relays

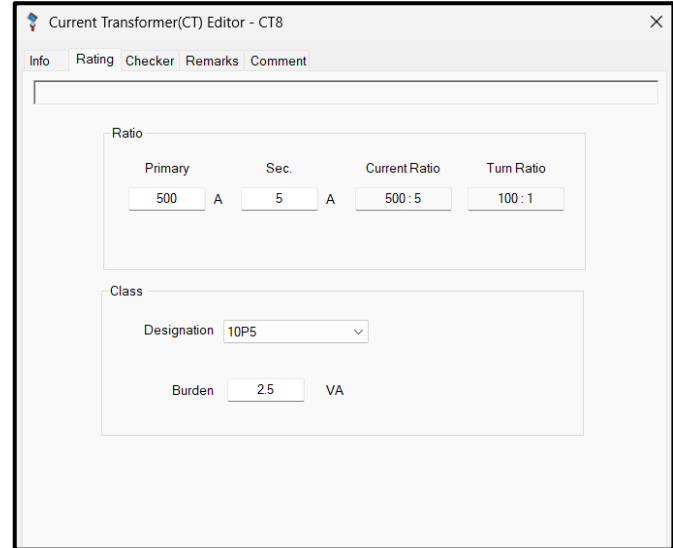
simulation

Current Transformer



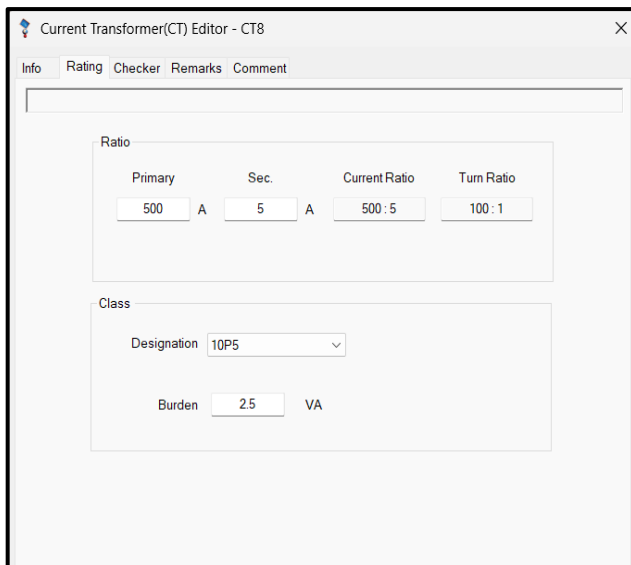
The screenshot shows the 'Current Transformer(CT) Editor - CT7' window. It has a tabbed interface with 'Info', 'Rating', 'Checker', 'Remarks', and 'Comment'. The 'Info' tab is active. Under the 'Ratio' section, the 'Primary' is set to 400 A, 'Sec.' is 5 A, 'Current Ratio' is 400:5, and 'Turn Ratio' is 80:1. Under the 'Class' section, the 'Designation' is 10P5 and the 'Burden' is 2.5 VA.

Figure 29:current transformer for Braker 4



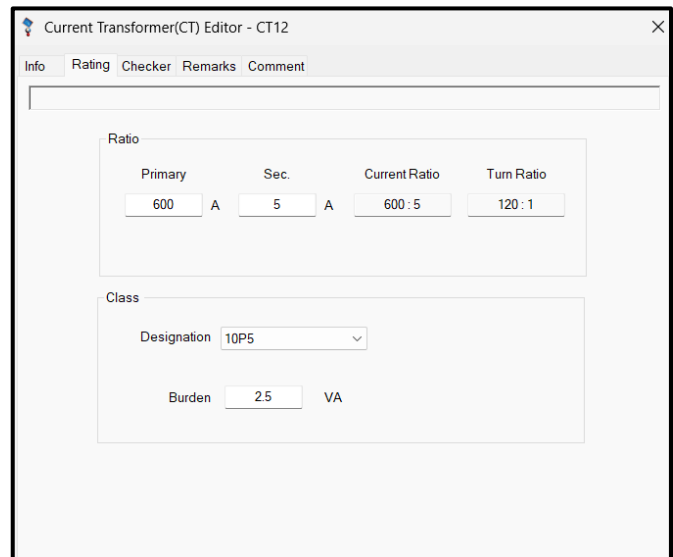
The screenshot shows the 'Current Transformer(CT) Editor - CT8' window. It has a tabbed interface with 'Info', 'Rating', 'Checker', 'Remarks', and 'Comment'. The 'Info' tab is active. Under the 'Ratio' section, the 'Primary' is set to 500 A, 'Sec.' is 5 A, 'Current Ratio' is 500:5, and 'Turn Ratio' is 100:1. Under the 'Class' section, the 'Designation' is 10P5 and the 'Burden' is 2.5 VA.

Figure 28:current transformer for Braker 3



The screenshot shows the 'Current Transformer(CT) Editor - CT8' window. It has a tabbed interface with 'Info', 'Rating', 'Checker', 'Remarks', and 'Comment'. The 'Info' tab is active. Under the 'Ratio' section, the 'Primary' is set to 500 A, 'Sec.' is 5 A, 'Current Ratio' is 500:5, and 'Turn Ratio' is 100:1. Under the 'Class' section, the 'Designation' is 10P5 and the 'Burden' is 2.5 VA.

Figure 31: current transformer for Braker2



The screenshot shows the 'Current Transformer(CT) Editor - CT12' window. It has a tabbed interface with 'Info', 'Rating', 'Checker', 'Remarks', and 'Comment'. The 'Info' tab is active. Under the 'Ratio' section, the 'Primary' is set to 600 A, 'Sec.' is 5 A, 'Current Ratio' is 600:5, and 'Turn Ratio' is 120:1. Under the 'Class' section, the 'Designation' is 10P5 and the 'Burden' is 2.5 VA.

Figure 30: current transformer for Braker 1

By using the hand calculation, the settings of the current transformer were used to be implemented and coordinated between the breaker, where the value of CT ratio is chosen depending on the reflection of the normal current on the relay, where it should not exceed 5A maximum

Relays

Figure 32: Relay of breaker 4

Figure 33: Relay of breaker 3

Figure 34: Relay of breaker 2

Figure 35: Relay of breaker 1

The

The relay was set to (CO8) from >>library>>> model (CO). After that, select the Type (Co8-invers). Each relay has TDs, CTs to be coordinated without any problem, and every relay takes its time to detect the fault and send the signal to the breaker to break the circuit. If it does not remove the fault after it takes its full time, the backup breaker will react to remove the fault

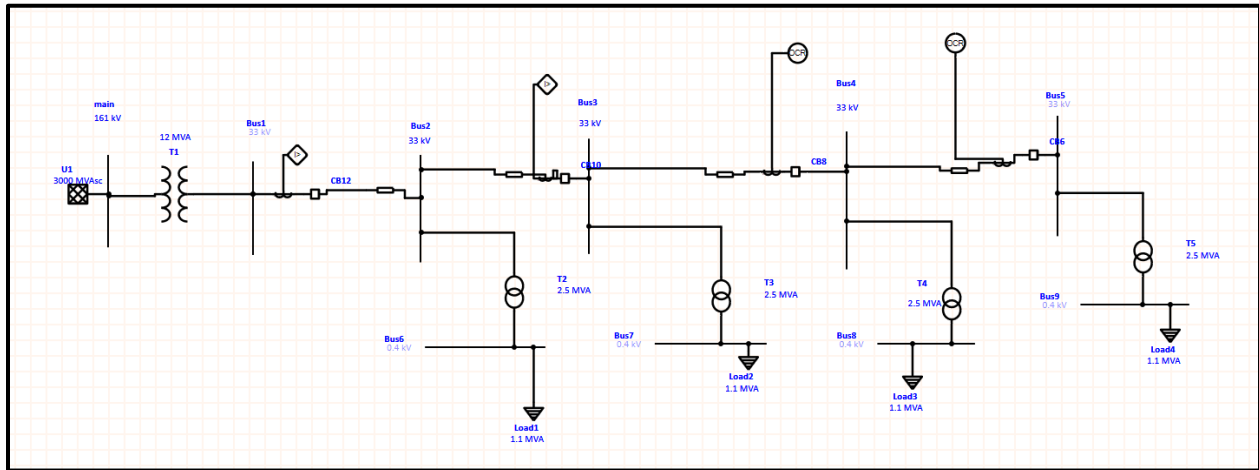


Figure 36: Connecting the relays

To make fault mode >>> (star-protection and coordination were used, where the setting of the relay was calculated and implemented in the circuit as the figures above shown

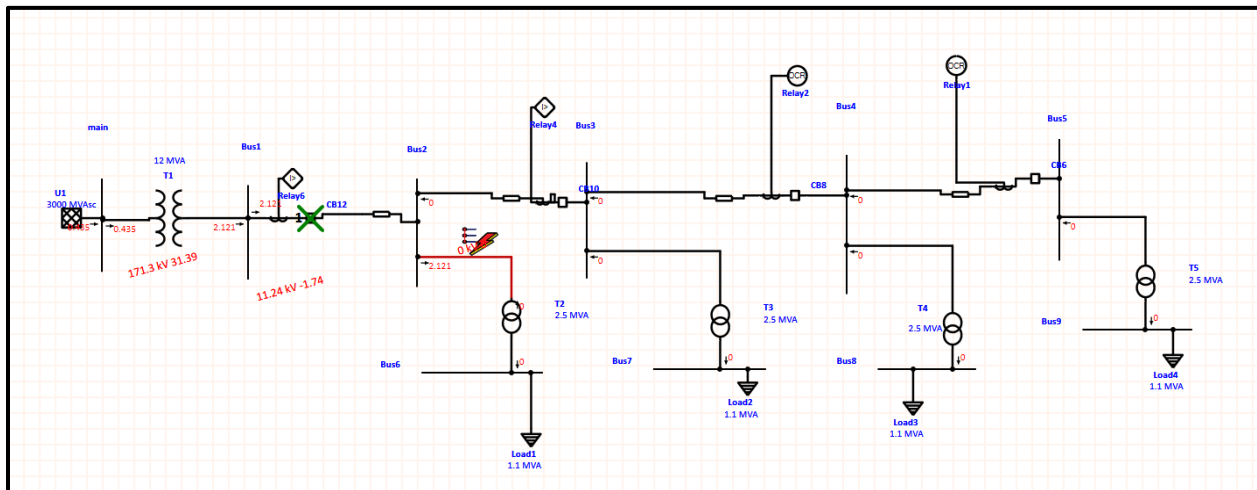


Figure 37: fault at bus 2

When a fault happens on the bus 2 the main breaker will see the fault and react to breaking the circuit while the other breaker will not detect the fault because the fault behind them

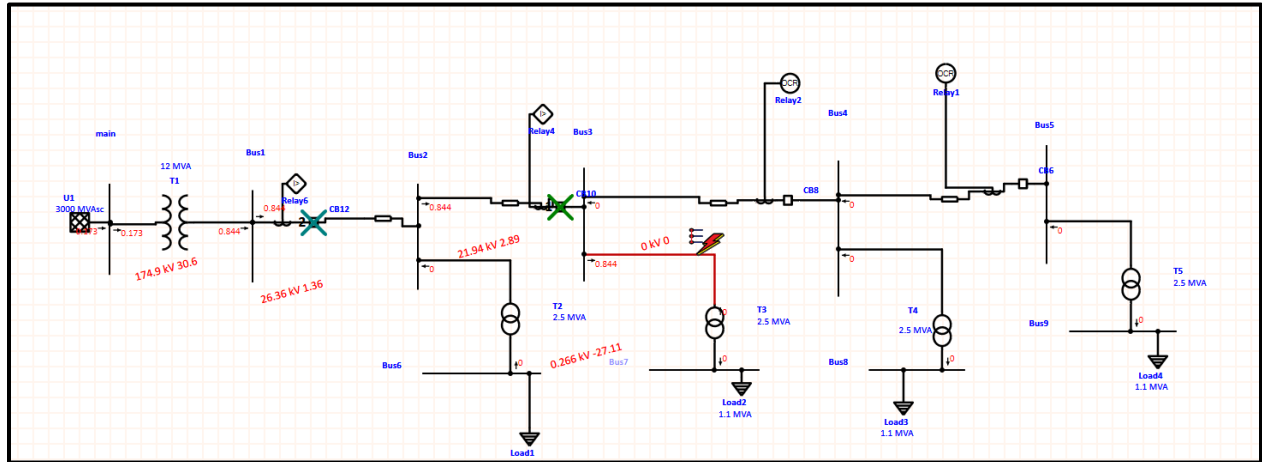


Figure 38: Fault at Bus 4

When the fault happens on Bus 3, the B3 will react if it is fall in its job to remove the fault after it is take all the time that needed to react for his fault the backup protection device (main breaker) will work as sup protective device and break the circuit

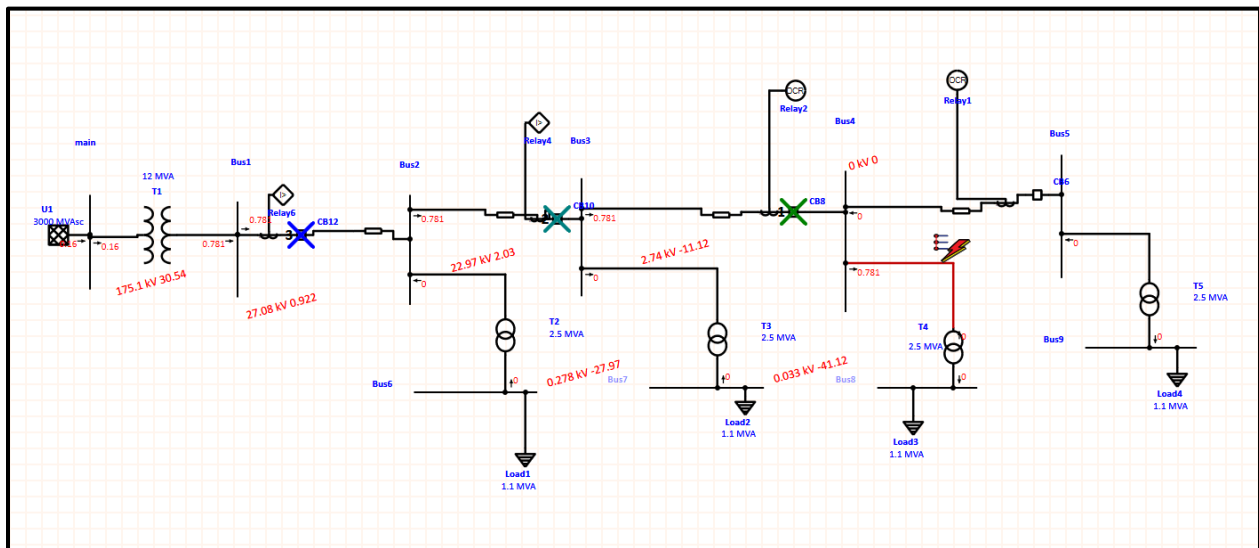


Figure 39 : Fault at bus3

When the fault occurs on bus 4, breaker 3 will react first. If it fails and after it takes the total time that need , the breaker B2, which means after 0.78sec, that is all the time that the circuit breaker B3 is needed to detect and break the circuit, this time includes (relay time + coordination time + 5 cycles for the breaker), will work after it, to turn off the circuit if it is fails to remove the fault the main breaker will react to protect the circuit and remove fault

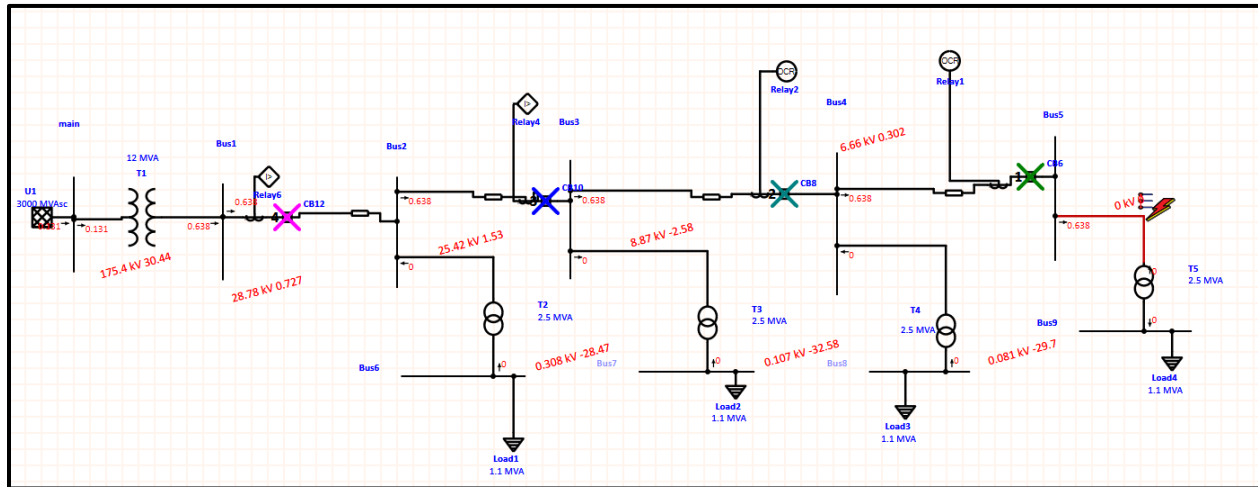


Figure 40: Fault at bus 5

When the fault occurs on bus 5 the breaker 4 will react first. If it fails, the breaker B3, which means after 0.433sec, that is all the time that the circuit breaker B3 is needed to detect and break the circuit, this time includes (relay time + coordination time + 5 cycles for the breaker) will work after it, to turn off the circuit when B3 is fail the B2 will works to turn off the circuit if B2 fail B1 will works

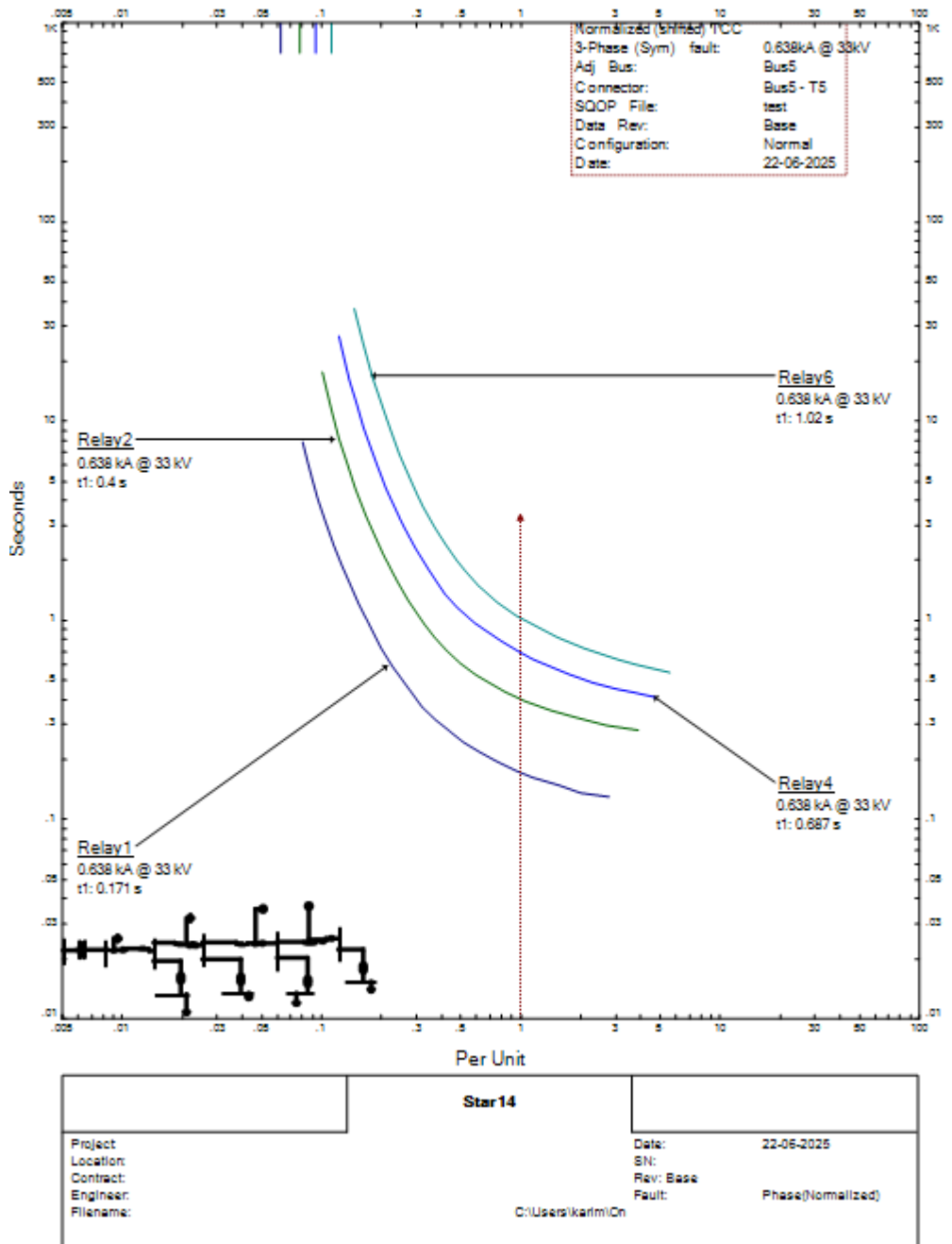


Figure 41: relays TDS curves

This graph shows the TDS time for the relay, it shows how they are coordinating together to make them work smoothly and without any problems where each breaker has its own current tap setting and current dial setting that was calculated to ensure no conflict between the relays and work properly to remove the fault in sequential no breaker react before the primary breaker take all the time needed to remove the fault it is important to notice the breaker that has the least time dial will react the faster and all the breaker react 20 times the pickup current when the fault current is two high

Phase 3

Part D : Fuses

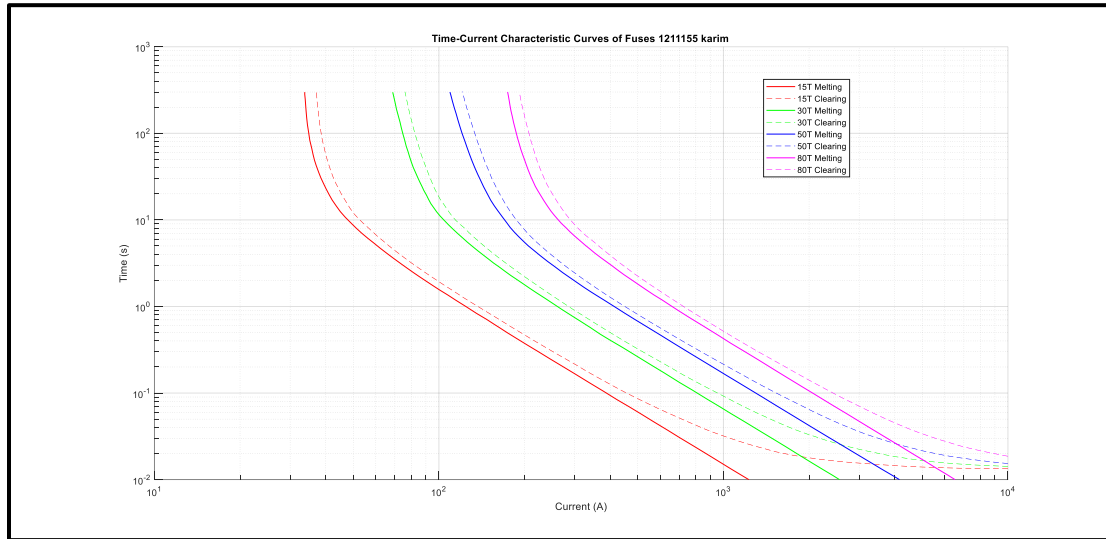


Figure 42: Time Melting, Time clearing for fuses

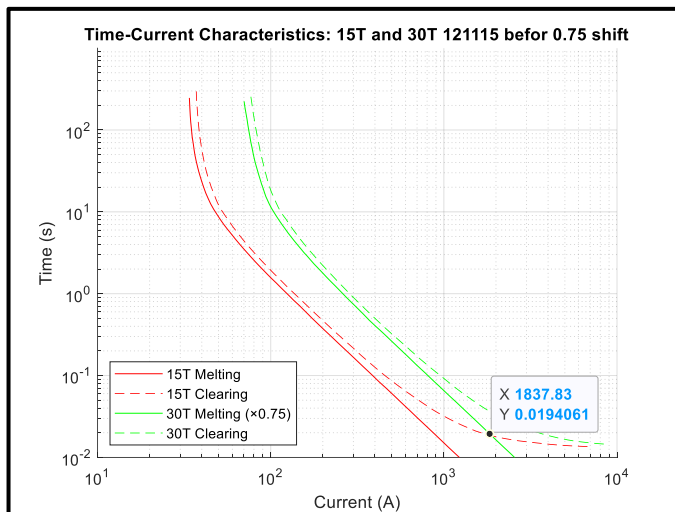


Figure 44: Before the Time-Current Curve Method

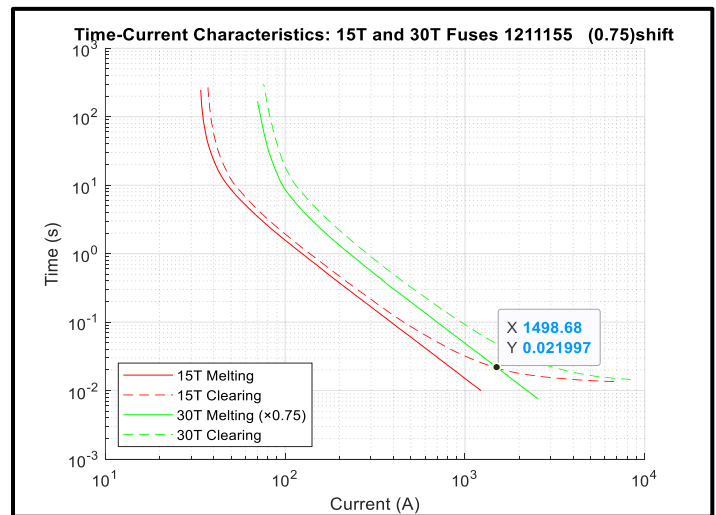


Figure 43: after The Time-Current Curve Method

The Time/Current Curve Method plots the minimum melting curve of the protected fuse against the total clearing curve of the protecting fuse. As a correction for pre-fault current and ambient temperature, the melting time is multiplied by 0.75 (the 75% rule). This gives satisfactory coordination, especially with the use of the same types of fuses.

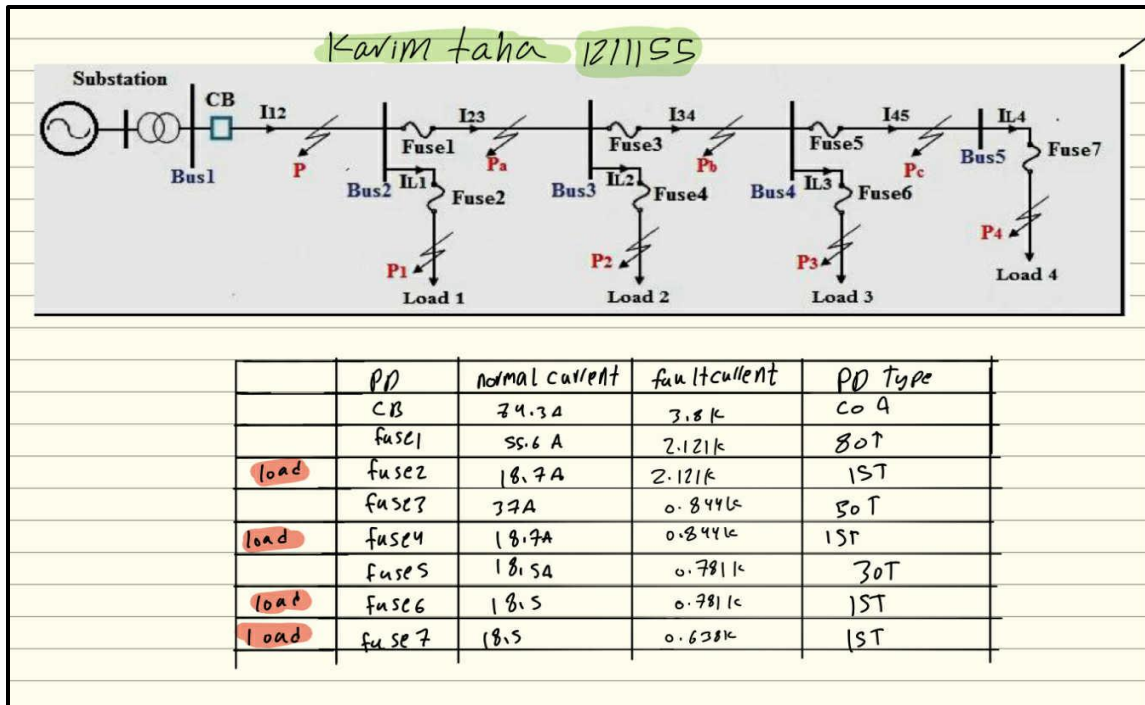


Figure 45: selection of fuses

To choose the fuses, we look for the normal current and the fault current for each fuse and do the necessary coordination

fuse 7: the normal current that gets thrown through it is 18.5A. From Table 3, T15 can handle normal current up to 23A

Fuse 5: The rated current passing through it is 18.5 A, and it must be coordinated with Fuse 7. It must also handle the fault current of 0.638 kA. Referring to Table 2, the appropriate fuse was T30.

Fuse 3: The normal current is 37 A, and it must operate with Fuse 5. It must also be able to handle a fault current of 0.844 kA. T50 was selected from Table 2 as it accommodates both the normal current.

Fuse 1: The rated current passing through it is 55.6A It must also be able to handle a fault current of 2.12K a. T50 was selected from Table 2 .T80 was chosen

Fuse 6,4, and 2 The normal current through them is 18.5 A, and they are automatically coordinated with the fuses on the transmission line.

Table 3: Ratings of fuses

Protecting	Protected link rating (amperes)													
Fuse Link	8T	10T	12T	15T	20T	25T	30T	40T	50T	65T	80T	100	140	200
Rating, A	Maximum fault current at which B will protect A (amperes)													
6T		350	600	920	1200	1500	2000	2540	3200	4100	5000	6100	9700	15.2
8T			375	800	1200	1500	2000	2540	3200	4100	5000	6100	9700	15.2
10T				530	1100	1500	2000	2540	3200	4100	5000	6100	9700	15.2
12T					680	1280	2000	2540	3200	4100	5000	6100	9700	15.2
15T						730	1700	2500	3200	4100	5000	6100	9700	15.2
20T							990	2100	3200	4100	5000	6100	9700	15.2
25T								1400	2600	4100	5000	6100	9700	15.2
30T									1500	3100	5000	6100	9700	15.2
40T										1700	3800	6100	9700	15.2
50T											1750	4400	9700	15.2
65T												2200	9700	15.2
80T													7200	15.2
100T													4000	13.8
140T														7.5

Table 4: current of fuses

TABLE 3.7 Continuous Current-Carrying Capacity of EEI-NEMA Fuse Links					
EEI-NEMA K or T Rating	Continuous Current (amperes)	EEI-NEMA K or T Rating	Continuous Current (amperes)	EEI-NEMA K or T Rating	Continuous Current (amperes)
6	9	20	30	65	95
8	12	25	38	80	120†
10	15	30	45	100	150†
12	18	40	60*	140	190
15	23	50	75*	200	200

* Only when used in a 100 or 200 ampere cutout.
† Only when used in a 200 ampere cutout.

These tables help with fuse coordination during fault operation and normal operation. It is important to choose a suitable fuse depending on the fault and the normal current that goes through the fuse with take a margin of increase in the load (normal current) because of an increase in population growth

Fuse Editor - Fuse1

Info | Rating | TCC kA | Model Info | Reliability | Checker | Remarks | Comment

S&C 27 kV max. 15T

Extra-Performance T Speed 5.3 kA

Standard

☒ ANSI

☐ IEC

Rating

kV	Size	Continuous Amp	Interrupting	Test PF
33	15T	15	5.3	8.3

Library...

Figure 46:Loda fuses

Fuse Editor - Fuse2

Info | Rating | TCC kA | Model Info | Reliability | Checker | Remarks | Comment

S&C 27 kV max. 30T

Extra-Performance T Speed 5.3 kA

Standard

☒ ANSI

☐ IEC

Rating

kV	Size	Continuous Amp	Interrupting	Test PF
33	30T	30	5.3	8.3

Library...

Figure 48:Transmission line fuse2

Fuse Editor - Fuse3

Info | Rating | TCC kA | Model Info | Reliability | Checker | Remarks | Comment

S&C 27 kV max. 50T

Extra-Performance T Speed 5.3 kA

Standard

☒ ANSI

☐ IEC

Rating

kV	Size	Continuous Amp	Interrupting	Test PF
33	50T	50	5.3	8.3

Library...

Figure 47:Transmission line fuse3

Fuse Editor - Fuse4

Info | Rating | TCC kA | Model Info | Reliability | Checker | Remarks | Comment

S&C 27 kV max. 80T

Extra-Performance T Speed 5.3 kA

Standard

☒ ANSI

☐ IEC

Rating

kV	Size	Continuous Amp	Interrupting	Test PF
33	80T	80	5.3	8.3

Library...

Figure 49:Transmission line fuse4

The fuses was selected from the library >Hierarchy >standard >ANSI>S&C> Extra performance T, The fuses were set at 33kV and the size was selected depending on the calculation that be done before

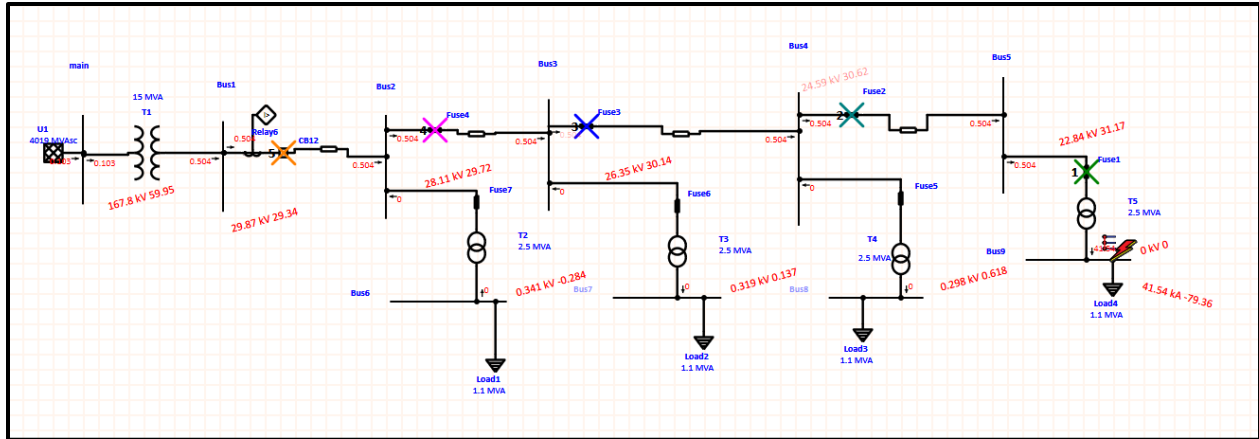


Figure 50: Fuses Fault Case 1

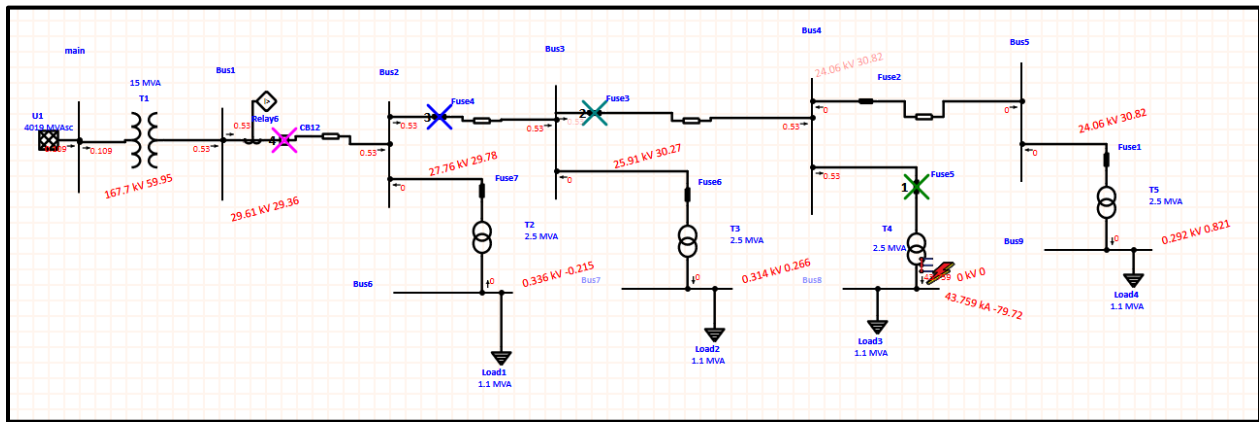


Figure 51: Fuses Fault Case2

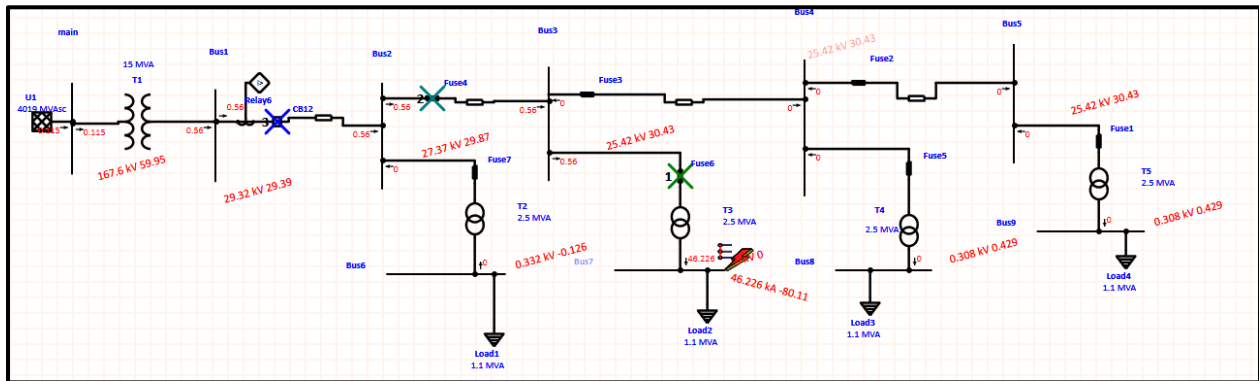


Figure 52: Fuses Fault Case3

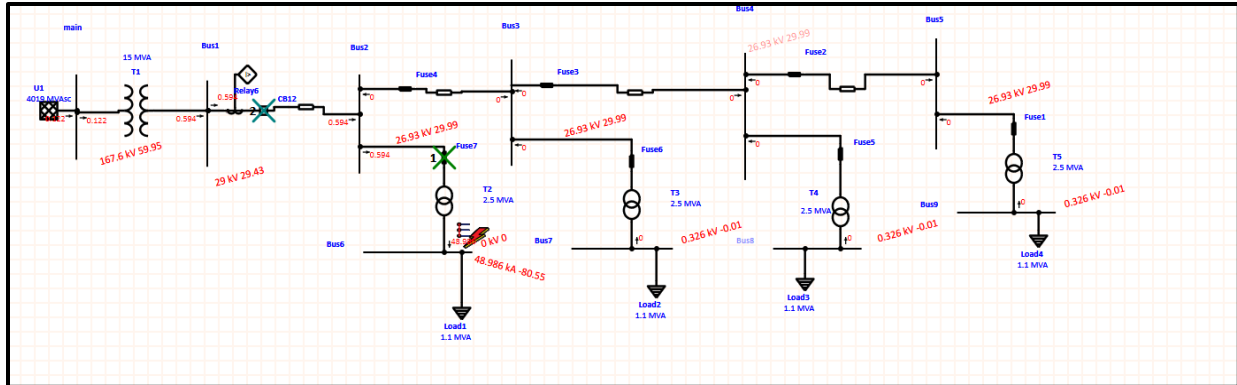


Figure 53:Fuses Fault Case4

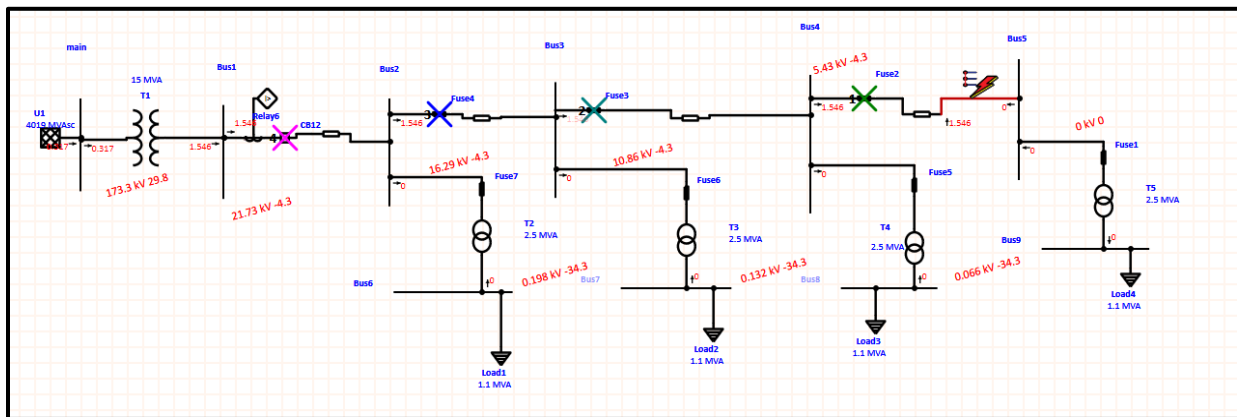


Figure 54:Fuses Fault Case5

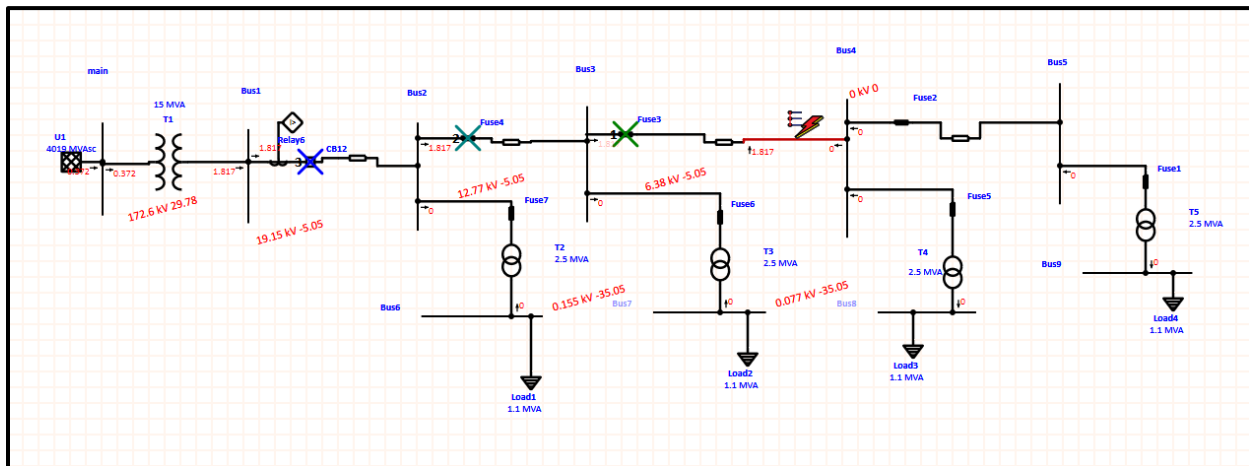


Figure 55:Fuses Fault Case6

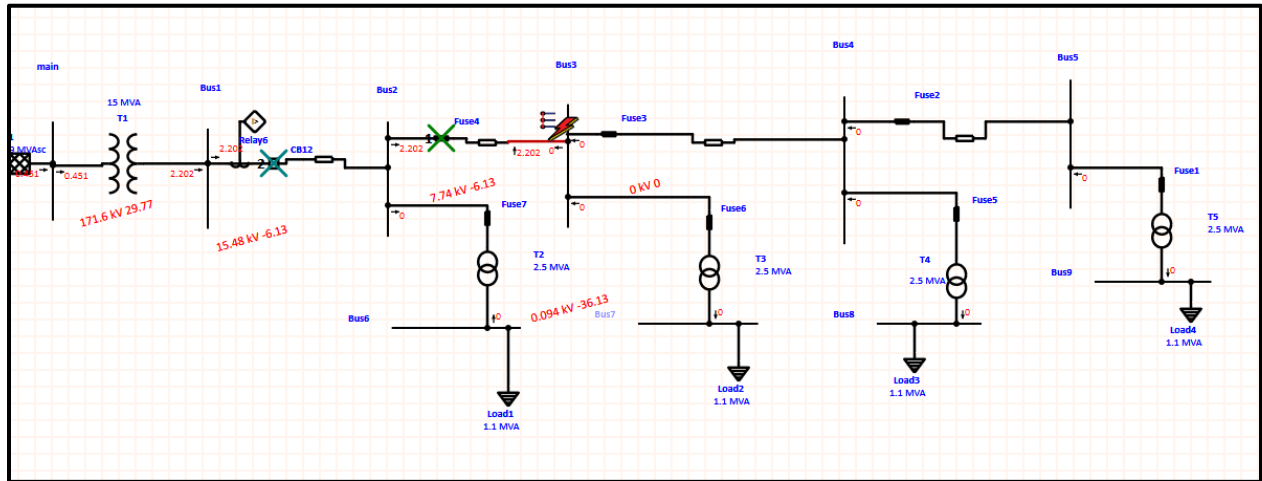


Figure 56:Fuses Fault Case6

Case 1: Fuse 1 will act as the primary protection device in case anything fails. Fuse 2 will be its backup in case Fuse 1 does not operate. Fuse 3 will also serve as a backup for Fuse 4. In case all the fuses fail, the circuit breaker will serve as the final protection.

Case 2: Fuse 5 will be the primary, and Fuse 3 will be the backup for Fuse 5 when it fails to operate when a fault occurs, and Fuse 1 backup for Fuse 3, Fuses 1 and 2 will not react to the fault.

Case 3: Fuse 6 will be the primary, and Fuse 4 will be the backup for Fuse 6 when it fails to operate when a fault happens. Finally, the main breaker will operate. Fuses (1,2,3and 5) will not operate

Case 4: Fuse 1 will be the primary, the main breaker will be the backup for Fuse 6 when it fails to operate when a fault. The other fuse will not see the fault

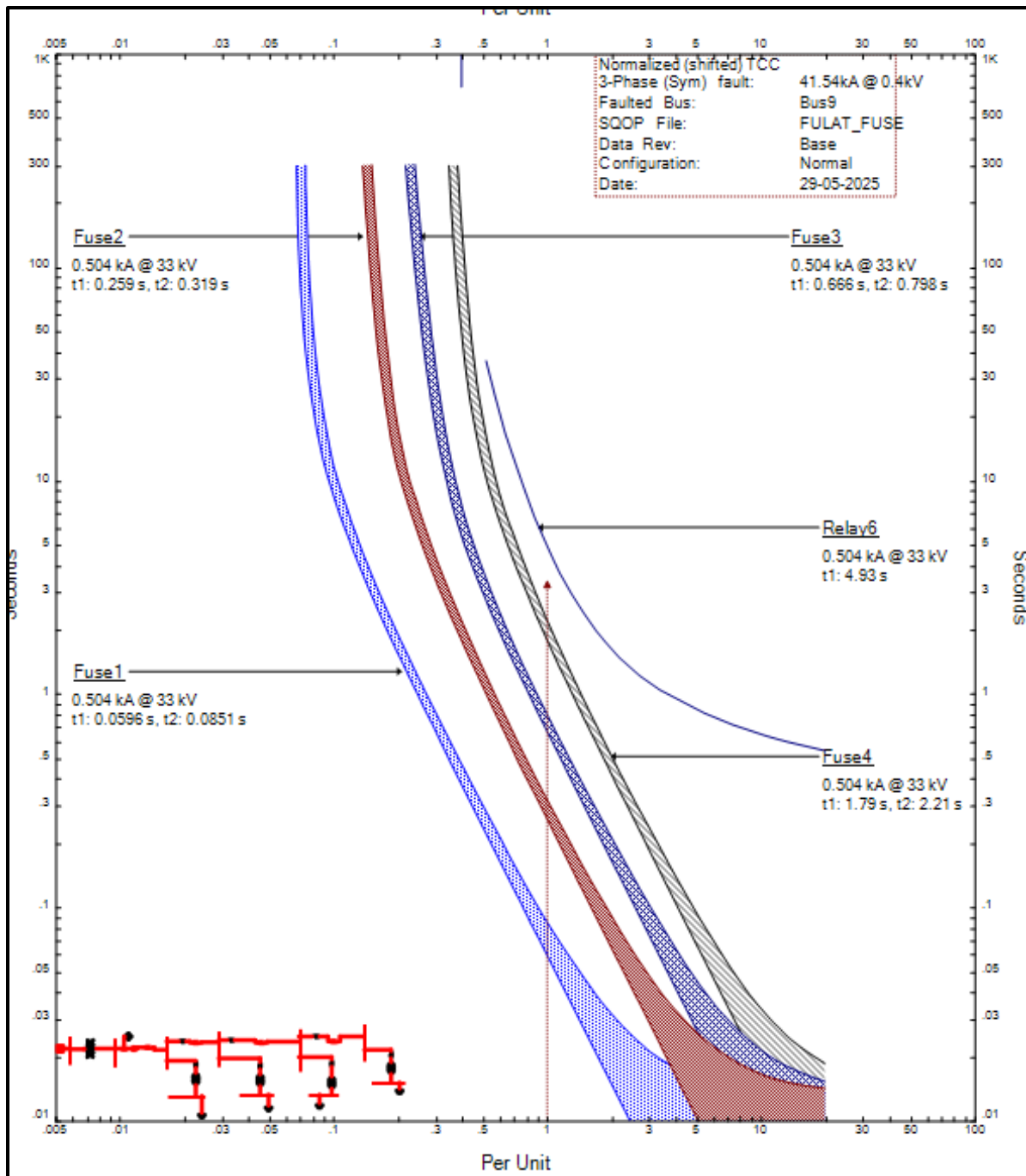


Figure 57: coordination between fuses

This curve shows the coordination between the fuses, where each fuse has a minimum melting time and total clearing time. Fuse 15T will be the primary protection device and the fastest fuse that operates when the fault appears at the load, then 30T, 50T, and 80T, where THE 80T fuse is the slowest of all the fuses, the circuit breaker will be the last protection device that operate when all fuses fail to clear the fault

Part E : Recloser

Figure 59: recloser settings 1

Figure 58:recloser settings 2

Figure 60: recloser settings 3

Figure 61:Fuses settings

To choose the recloser type, go to >controller >controller lib> choice cooper>form 4C the ratings were selected depending on the value of the network, and the coordination was done to make the recloser work as a backup for the fuse when temporary or permanent faults occur

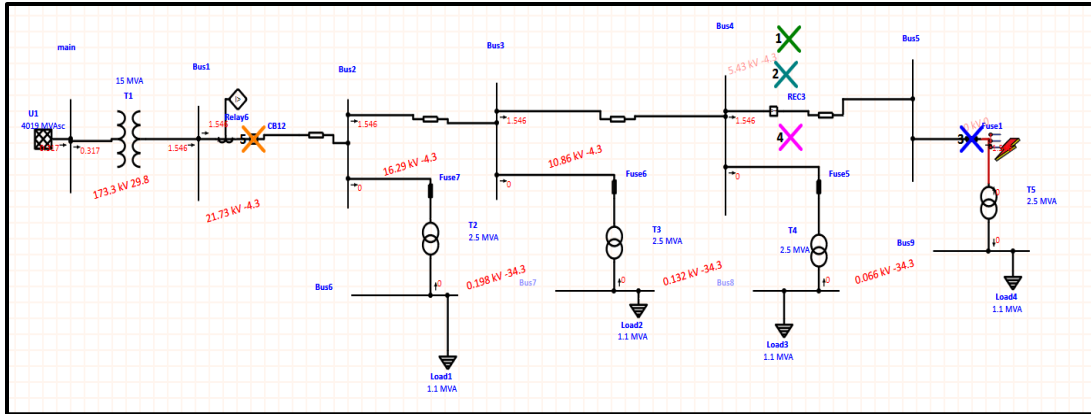


Figure 62:Recloser case 1

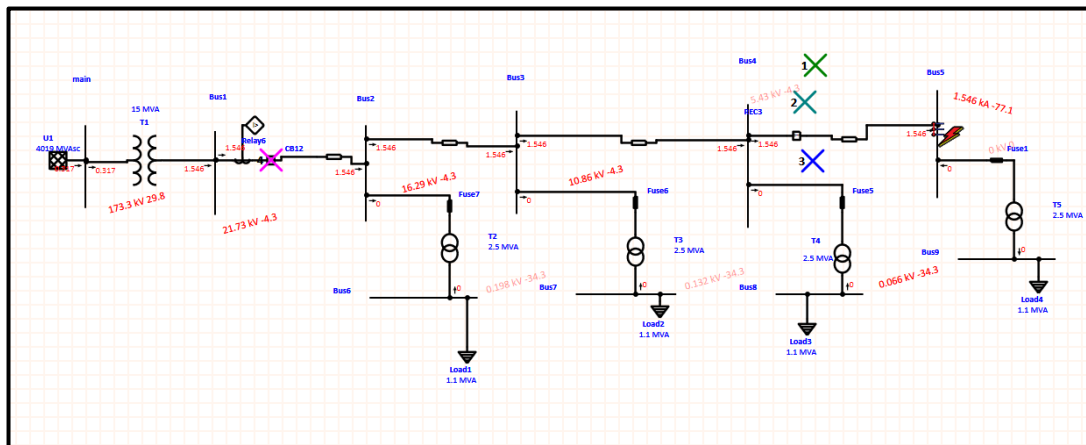


Figure 63: recloser case 2

Case 1: The recloser will work when the fault shows for the first time, and it breaks and reconnects when the fault is not cleared. The recloser will break again and reconnect if the fault does not appear, the fuses will melt since this is a permanent fault, not a temporary fault, and the recloser will work as a backup for the fuses if the fault is not removed after the fuse melts, and the circuit breaker will be the backup for the recloser

Case 2: When a fault occurs, the recloser will be the primary protection device and break twice for some time, and reconnect if the fault is not removed after that, it breaks for a third time, and the circuit breaker will be the backup for the recloser

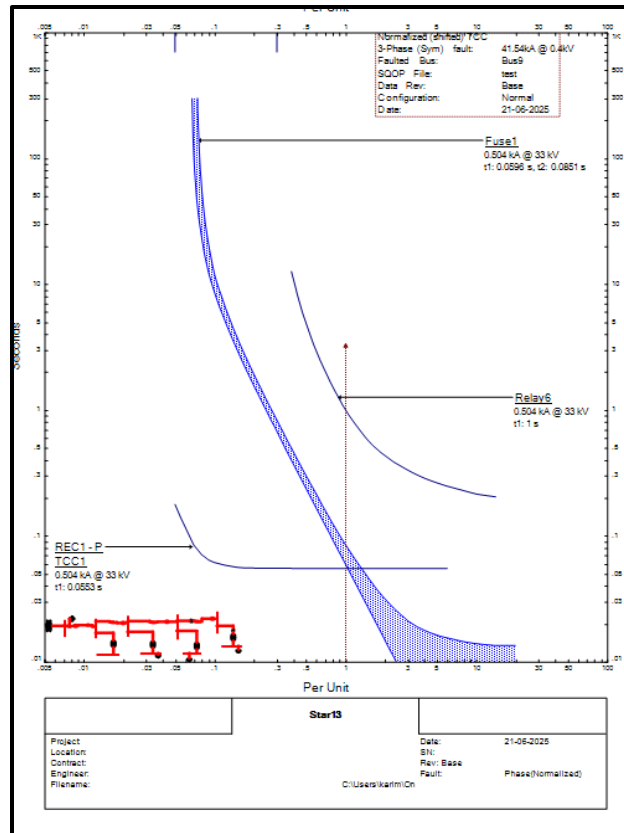


Figure 64:recloser coordination

This graph shows the coordination between the components (recloser, fuse, and circuit breaker) When a fault occurs the recloser will react for two time; this is in case temporary fault if the fault not clear the fuse will works if the fault still not clear the recloser will react this the FFS (Fast - Fast -Slow))for the recloser where it work as backup for the fuse and the circuit breaker works backup for the recloser

Conclusion

The skills of designing a circuit on ETAP were observed, and the effect of the transformer on the network and see the factors that the voltage drop and fault current depend on, as well as the knowledge about the current transformer and relays and how they coordinate with each other to break the circuit when the fault is occurred was observed, the function of fuses and the coordination between the fuses were examined and the function of the recloser and its function FFS(Fast-Fast-Slow) and the sequences of operation when the fault occurs was seen and discussed

References

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